

27	D	
28	C	$E = \frac{1}{2} m v^2 = \frac{p^2}{2m} \quad \square$ $p = (2mE)^{1/2} \quad (\text{m of alpha particle} = 4u)$ $= (8uE)^{1/2}$ $\lambda = h / p = h / (8uE)^{1/2} = h / 2(2uE)^{1/2}$
29	D	α will not pass through paper β is stopped by a few cm of aluminium foil only γ can pass through thick steel
30	B	$P = \frac{E}{t} = \frac{mc^2}{t}$ $= \frac{(6.0 \times 10^{-6})(3.00 \times 10^8)^2}{60 \times 60}$ $= 1.5 \times 10^8 \text{ W}$